Chapter 1

Introduction to IoT

Internet of Things (IoT) is a massive network of physical devices embedded with sensors, software, electronics, and network which allows the devices to exchange or collect data and perform certain actions.

Simply put, IoT is made up of two words: **Internet** & **Things**.

- Things – physical devices, appliances, gadgets, etc.
- Internet – through which these devices are connected

IoT aims at extending internet connectivity beyond computers and smartphones to other devices people use at home or for business. The technology allows devices to get controlled across network infrastructure remotely. As a result, it cuts down human effort and paves the way for accessing the connected devices easily. With autonomous control, the devices are operable without involving human interaction. IoT makes things virtually smart through AI algorithms, data collection, and networks enhancing our lives.

Examples: Pet tracking devices, diabetes monitors, AC sensors to adjust the temperature based on the outside temperature, smart wearables, and more.

IoT comprises things that have unique identities and are connected to internet. By 2020 there will be a total of 50 billion devices /things connected to internet. IoT is not limited to just connecting things to the internet but also allows things to communicate and exchange data.

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Definition: A dynamic global n/w infrastructure with self -configuring capabilities based on standard and interoperable communication protocols where physical and virtual —things‖ have identities, physical attributes and virtual personalities and use intelligent interfaces, and are seamlessly integrated into information n/w, often communicate data associated with users and their environments.

Characteristics of IoT

1) **Dynamic & Self Adapting:** IoT devices and systems may have the capability to dynamically adapt with the changing contexts and take actions based on their operating conditions, user‘s context or sensed environment. Eg: the surveillance system is adapting itself based on context and changing conditions.

2) **Self Configuring:** allowing a large number of devices to work together to provide certain functionality.

3) **Inter Operable Communication Protocols:** support a number of interoperable communication protocols and can communicate with other devices and also with infrastructure.

4) **Unique Identity:** Each IoT device has a unique identity and a unique identifier (IP address).

5) **Integrated into Information Network:** that allow them to communicate and exchange data with other devices and systems.

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IoT Categories

1. ***Consumer IoT (CIoT)*** refers to the use of IoT for consumer applications and devices. Common CIoT products include smartphones, wearables, smart assistants, home appliances, etc. Typically, CIoT solutions leverage Wi-Fi, Bluetooth, and ZigBee to facilitate connectivity. These technologies offer short-range communication suitable for deployments in smaller venues, such as homes and offices.

While CIoT tends to focus on augmenting personal and home environments, Commercial IoT goes a bit further, delivering the benefits of IoT to larger venues. Think: commercial office buildings, supermarkets, stores, hotels, healthcare facilities, and entertainment venues.

There are numerous use cases for commercial IoT, including monitoring environmental conditions, managing access to corporate facilities, and economizing utilities and consumption in hotels and other large venues. Many Commercial IoT solutions are geared towards improving customer experiences and business conditions.

2. **Industrial IoT (IIoT),** is perhaps the most dynamic wing of the IoT industry. Its focus is on augmenting existing industrial systems, making them both more productive and more efficient. IIoT deployments are typically found in large-scale factories and manufacturing plants and are often associated with industries like healthcare, agriculture, automotive, and logistics. The Industrial Internet is perhaps the most well-known example of IIoT.

3. ***Infrastructure IoT*** is concerned with the development of smart infrastructures that incorporate IoT technologies to boost efficiency, cost savings, maintenance, etc. This includes the ability to monitor and control operations of urban and rural infrastructures, such as bridges, railway tracks, and on- and offshore windfarms. Technically speaking, infrastructure IoT is a subset of IIoT. However, due to its significance, it's often treated as its own separate thing.
4. The last type of IoT is the ***Internet of Military Things (IoMT)***, often referred to as Battlefield IoT, the Internet of Battlefield Things, or simply IoBT. IoMT is precisely what it sounds like — the use of IoT in military settings and battlefield situations. It is chiefly aimed at increasing situational awareness, bolstering risk assessment, and improving response times. Common IoMT applications include connecting ships, planes, tanks, soldiers, drones, and even Forward Operating Bases via an interconnected system. In addition, IoMT produces data that can be leveraged to improve military practices, systems, equipment, and strategy.

IoT Enablers and Connectivity Layers

System installers, repairers, craftsmen, electricians, plumbers, architects who connect devices and systems to the Internet for personal use and for commercial and other business uses.

As the Internet of Things (IoT) enables devices to make intelligent decisions that generate positive business outcomes, it's the sensors that enable those decisions. As cost and time-to-market pressures continue to rise, sensors provide greater visibility into connected systems and empower those systems to react intelligently to changes driven by both external forces and internal factors.

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Sensors are the components that provide the actionable insights that power the IoT and enable organizations to make more effective business decisions. It's through this real-time measurement that the IoT can transform an organization's ability to react to change.

Wi-Fi was designed for computers, and 4G LTE wireless targeted smartphones and portable devices. Both have been tremendously successful — and both were shaped by the devices they were intended for. The same goes for 5G, the first generation of wireless technology designed with extremely small, low-power, and near-ubiquitous IoT devices in mind.

Unlike Wi-Fi and LTE devices, which we handle and plug into power sources on a daily basis, IoT sensors will operate autonomously for years at a time, often in inaccessible places, without recharging or replacement. An explosion of new protocols: The IoT is prompting the development of several different 5G communication standards, not just one or two network types

Baseline Technologies of IoT

According to Jones, the top 10 emerging IoT technologies are:

1. IoT Security: Security technologies will be required to protect IoT devices and platforms from both attacks and physical tampering, to encrypt their communications, and to address new challenges such as impersonating "things" or denial-of-sleep attacks that drain batteries. IoT security will be complicated by the fact that many "things" use simple processors and operating systems that may not support sophisticated security approaches.

2. IoT Analytics: IoT business models will exploit the information collected by "things" in many ways, which will demand new analytic tools and algorithms. As data volumes increase over the next five years, the needs of the IoT may diverge further from traditional analytics.

3. IoT Device (Thing) Management: Long-lived nontrivial "things" will require management and monitoring, including device monitoring, firmware and software updates, diagnostics, crash analysis and reporting, physical management, and security management. Tools must be capable of managing and monitoring thousands and perhaps even millions of devices.

4. Low-Power, Short-Range IoT Networks. Low-power, short-range networks will dominate wireless IoT connectivity through 2025, far outnumbering connections using wide-area IoT networks. However, commercial and technical trade-offs mean that many solutions will coexist, with no single dominant winner.

5. Low-Power, Wide-Area Networks. Traditional cellular networks don't deliver a good combination of technical features and operational cost for those IoT applications that need wide-area coverage combined with relatively low bandwidth, good battery life, low hardware and operating cost, and high connection density. Emerging standards such as narrowband IoT will likely dominate this space.

6. IoT Processors. The processors and architectures used by IoT devices define many of their capabilities, such as whether they are capable of strong security and encryption, power consumption, whether they are sophisticated enough to support an operating system, updatable firmware, and embedded device management agents. Understanding the implications of processor choices will demand deep technical skills.

7. IoT Operating Systems. Traditional operating systems such as Windows and iOS were not designed for IoT applications. They consume too much power, need fast processors, and in some cases, lack features such as guaranteed real-time response. They also have too large a memory footprint for small devices and may not support the chips that IoT developers use. Consequently, a wide range of IoT-specific operating systems has been developed to suit many different hardware footprints and feature needs.

8. Event Stream Processing: Some IoT applications will generate extremely high data rates that must be analyzed in real time. Systems creating tens of thousands of events per second are common, and millions of events per second can occur in some situations. To address such requirements, distributed stream computing platforms have emerged that can process very high-rate data streams and perform tasks such as real-time analytics and pattern identification.

9. IoT Platforms. IoT platforms bundle many of the infrastructure components of an IoT system into a single product. The services provided by such platforms fall into three main categories:

Low-level device control and operations such as communications, device monitoring and management, security, and firmware updates; IoT data acquisition, transformation and management; IoT application development, including event-driven logic, application programming, visualization, analytics and adapters to connect to enterprise systems.

10. IoT Standards and Ecosystems. Standards and their associated **application programming interfaces (APIs)** will be essential because IoT devices will need to interoperate and communicate, and many IoT business models will rely on sharing data between multiple devices and organizations. Many IoT ecosystems will emerge, and organizations creating products may have to develop variants to support multiple standards or ecosystems and be prepared

to update products during their life span as the standards evolve and new standards and APIs emerge.

Sensors

Sensors are used for sensing things and devices etc. A sensor is a device that provides ausable output in response to a specified measurement. The sensor attains a physical parameter and converts it into a signal suitable for processing (e.g. electrical, mechanical, optical) the characteristics of any device or material to detect the presence of a particular physical quantity. The output of the sensor is a signal which is converted to a human-readable form like changes in characteristics, changes in resistance, capacitance, impedance etc. here are myriad different sensors available to measure virtually everything in the physical world. There are a number of ways to group and cluster sensors into different categories, including the following:

■ Active or passive: Sensors can be categorized based on whether they produce an energy output and typically require an external power supply (active) or whether they simply receive energy and typically require no external power supply (passive).

■ Invasive or non-invasive: Sensors can be categorized based on whether a sensor is part of the environment it is measuring (invasive) or external to it (non-invasive).

■ Contact or no-contact: Sensors can be categorized based on whether they require physical contact with what they are measuring (contact) or not (no-contact).

■ Absolute or relative: Sensors can be categorized based on whether they measure on an absolute scale (absolute) or based on a difference with a fixed or variable reference value (relative).

■Area of application: Sensors can be categorized based on the specific industry or vertical where they are being used.

■ How sensors measure: Sensors can be categorized based on the physical mechanism used to measure sensory input (for example, thermoelectric, electrochemical, piezo resistive, optic, electric, fluid mechanic, photoelastic).

■ What sensors measure: Sensors can be categorized based on their applications or what physical variables they measure.

Characteristics of a Sensor

The static accuracy of a sensor indicates how much the sensor signal correctly represents the measured quantity after it stabilizes (i.e. beyond the transient period.) Important **static characteristics** of sensors include **sensitivity, resolution, linearity, zero drift and full-scale drift, range, repeatability and reproducibility.**

1. ***Sensitivity*** is a measure of the change in output of the sensor relative to a unit change in the input (the measured quantity.) Example: The speakers you purchase for your home entertainment may have a rated sensitivity of 89 dB Signal Pressure Level per Watt per meter.

2. ***Resolution*** is the smallest amount of change in the input that can be detected and accurately indicated by the sensor. Example: What is the resolution of an ordinary ruler? of a Vernier Calipers?

3. ***Linearity*** is determined by the calibration curve. The static calibration curve plots the output amplitude versus the input amplitude under static conditions. Its degree of resemblance to a straight line describes the linearity.

4. ***Drift*** is the deviation from a specific reading of the sensor when the sensor is kept at that value for a prolonged period of time. The ***zero drift*** refers to the change in sensor output if the input is kept steady at a level that (initially) yields a zero reading.

   Similarly, the ***full -scale drift*** is the drift if the input is maintained at a value which originally yields a full scale deflection. Reasons for drift may be extraneous, such as changes in ambient pressure, humidity, temperature etc., or due to changes in the constituents of the sensor itself, such as aging, wear etc.

5. The ***range*** of a sensor is determined by the allowed lower and upper limits of its input or output. Usually, the range is determined by the accuracy required.

6. ***Repeatability*** is defined as the deviation between measurements in a sequence when the object under test is the same and approaches its value from the same direction each time. The measurements have to be made under a short enough time duration so as not to allow significant long term drift. Repeatability is usually specified as a percentage of the sensor range.

7. ***Reproducibility*** is the same as repeatability, except it also incorporates long time lapses between subsequent measurements. The sensor has to be operation between measurements but must be calibrated. **Reproducibility** is specified as a percentage of the sensor range per unit of time.

The **dynamic characteristics** of a sensor represent the time response of the sensor system. Knowledge of these is essential to fruitfully use a sensor. Important common dynamic responses of sensors include rise time, delay time, peak time, settling time percentage error and steady-state error

Sensor Types	Description	Examples
Position	A position sensor measures the position of an object; the position measurement can be either in absolute terms (absolute position sensor) or in relative terms (displacement sensor). Position sensors can be linear, angular, or multi-axis.	Potentiometer, inclinometer, proximity sensor
Occupancy and motion	Occupancy sensors detect the presence of people and animals in a surveillance area, while motion sensors detect movement of people and objects. The difference between the two is that occupancy sensors generate a signal even when a person is stationary, whereas motion sensors do not.	Electric eye, radar
Velocity and acceleration	Velocity (speed of motion) sensors may be linear or angular, indicating how fast an object moves along a straight line or how fast it rotates. Acceleration sensors measure changes in velocity.	Accelerometer, gyroscope
Force	Force sensors detect whether a physical force is applied and whether the magnitude of force is beyond a threshold.	Force gauge, viscometer, tactile sensor (touch sensor)
Pressure	Pressure sensors are related to force sensors, measuring force applied by liquids or gases. Pressure is measured in terms of force per unit area.	Barometer, Bourdon gauge, piezometer
Flow	Flow sensors detect the rate of fluid flow. They measure the volume (mass flow) or rate (flow velocity) of fluid that has passed through a system in a given period of time.	Anemometer, mass flow sensor, water meter

Sensor Types	Description	Examples
Acoustic	Acoustic sensors measure sound levels and convert that information into digital or analog data signals.	Microphone, geophone, hydrophone
Humidity	Humidity sensors detect humidity (amount of water vapor) in the air or a mass. Humidity levels can be measured in various ways: absolute humidity, relative humidity, mass ratio, and so on.	Hygrometer, humistor, soil moisture sensor
Light	Light sensors detect the presence of light (visible or invisible).	Infrared sensor, photodetector, flame detector
Radiation	Radiation sensors detect radiation in the environment. Radiation can be sensed by scintillating or ionization detection.	Geiger-Müller counter, scintillator, neutron detector
Temperature	Temperature sensors measure the amount of heat or cold that is present in a system. They can be broadly of two types: contact and non-contact. Contact temperature sensors need to be in physical contact with the object being sensed. Non-contact sensors do not need physical contact, as they measure temperature through convection and radiation.	Thermometer, calorimeter, temperature gauge
Chemical	Chemical sensors measure the concentration of chemicals in a system. When subjected to a mix of chemicals, chemical sensors are typically selective for a target type of chemical (for example, a CO_2 sensor senses only carbon dioxide).	Breathalyzer, olfactometer, smoke detector
Biosensors	Biosensors detect various biological elements, such as organisms, tissues, cells, enzymes, antibodies, and nucleic acid.	Blood glucose biosensor, pulse oximetry, electrocardiograph

Classification of Sensors

The common IoT sensors include:

Temperature sensors, Pressure sensors, Motion sensors, Level sensors, Image sensors, Proximity sensors, Water quality sensors, Chemical sensors, Gas sensors, Smoke sensors, Infrared (IR) sensors, Humidity sensors, etc.

A description of each of these sensors is provided below.

Temperature sensors

Temperature sensors detect the temperature of the air or a physical object and concert that temperature level into an electrical signal that can be calibrated accurately reflect the measured temperature. These sensors could monitor the temperature of the soil to help with agricultural output or the temperature of a bearing operating in a critical piece of equipment to sense when it might be overheating or nearing the point of failure.

Pressure sensors

Pressure sensors measure the pressure or force per unit area applied to the sensor and can detect things such as atmospheric pressure, the pressure of a stored gas or liquid in a sealed system such as tank or pressure vessel, or the weight of an object.

Motion sensors

Motion sensors or detectors can sense the movement of a physical object by using any one of several technologies, including passive infrared (PIR), microwave detection, or ultrasonic, which uses sound to detect objects. These sensors can be used in security and intrusion detection systems, but can also be used to automate the control of doors, sinks, air conditioning and heating, or other systems.

Level sensors

Level sensors translate the level of a liquid relative to a benchmark normal value into a signal. Fuel gauges display the level of fuel in a vehicle's tank, as an example, which provides a continuous level reading. There are also point level sensors, which are a go-no/go or digital representation of the level of the liquid. Some automobiles have a light that illuminates when the fuel level tank is very close to empty, acting as an alarm that warns the driver that fuel is about to run out completely.

Image sensors

Image sensors function to capture images to be digitally stored for processing. License plate readers are an example, as well as facial recognition systems. Automated production lines can use image sensors to detect issues with quality such as how well a surface is painted after leaving the spray booth.

Proximity sensors

Proximity sensors can detect the presence or absence of objects that approach the sensor through a variety of different technology designs.

Water quality sensors

The importance of water to human beings on earth not only for drinking but as a key ingredient needed in many production processes dictates the need to be able to sense and measure parameters around water quality. Some examples of what is sensed and monitored include:

Chemical presence (such as chlorine levels or fluoride levels),Oxygen levels (which may impact the growth of algae and bacteria),Electrical conductivity (which can indicate the level of ions present in water), pH level (a reflection of the relative acidity or alkalinity of the water),Turbidity levels (a measurement of

the amount of suspended solids in water)

Chemical sensors

Chemical sensors are designed to detect the presence of specific chemical substances which may have inadvertently leaked from their containers into spaces that are occupied by personnel and are useful in controlling industrial process conditions.

Gas sensors

Related to chemical sensors, gas sensors are tuned to detect the presence of combustible, toxic, or flammable gas in the vicinity of the sensor. Examples of specific gases that can be detected include:

Bromine (Br_2), Carbon Monoxide (CO), Chlorine (Cl_2), Chlorine Dioxide (ClO_2),Hydrogen Cyanide (HCN),Hydrogen Peroxide (H_2O_2), Hydrogen Sulfide (H_2S), Nitric Oxide (NO), Nitrogen Dioxide (NO_2), Ozone (O_3), etc.

Smoke sensors

Smoke sensors or detectors pick up the presence of smoke conditions which could be an indication of a fire typically using optical sensors (photoelectric detection) or ionization detection.

Infrared (IR) sensors

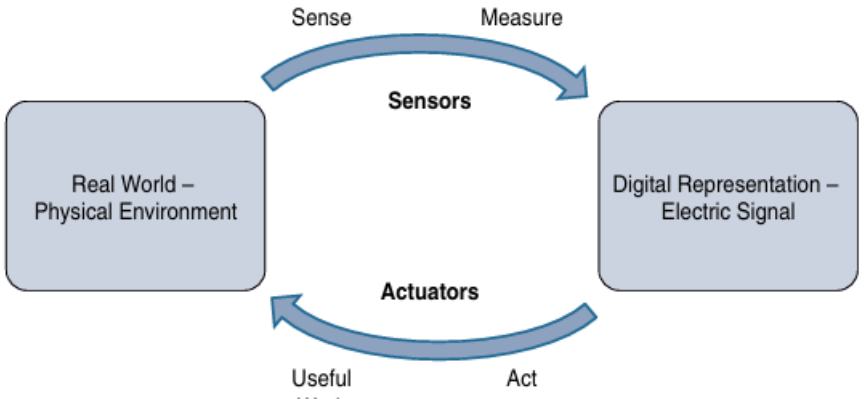
Infrared sensor technologies detect infrared radiation that is emitted by objects. Non-contact thermometers make use of these types of sensors as a way of measuring the temperature of an object without having to directly place a probe or sensor on that object. They find use in analyzing the heat signature of electronics and detecting blood flow or blood pressure in patients.

Acceleration sensors

While motion sensors detect movement of an object, acceleration sensors, or accelerometers as they are also known, detect the rate of change of velocity of an object. This change may be due to a free-fall condition, a sudden vibration that is causing movement with speed changes, or rotational motion (a directional change).

Actuators

An actuator is a machine component or system that moves or controls the mechanism or the system. Sensors in the device sense the environment, then control signals are generated for the actuators according to the actions needed to perform. Actuators convert an electrical signal into a corresponding physical quantity such as movement, force, sound etc. Figure demonstrates the symmetry and complementary nature of these two types of devices.



Types of Actuators

1. **Servo Motors:**

Servo is a small device that incorporates a two wire DC motor, a gear train, a potentiometer, an integrated circuit, and a shaft (output spine).



2. **Stepper Motors:**



Stepper motors are DC motors that move in discrete steps. They have multiple coils that are organized in groups called “phases”. By energizing each phase in sequence, the motor will rotate, one step at a time. With a computer controlled stepping, you can achieve very precise positioning and/or speed control.

3. **DC Motors (Continuous Rotation Motors):**



Direct Current (DC) motor is the most common actuator used in projects. They are simple, cheap, and easy to use. DC motors convert electrical into mechanical energy. Also, they come in different sizes.

4. **Linear actuator:**

A linear actuator is an actuator that creates motion in a straight line, in contrast to the circular motion of a conventional electric motor. Linear actuators are used in machine tools and industrial machinery, in computer peripherals such as disk drives and printers, in valves and dampers, and in many other places where linear motion is required.



5. **Relay:**

relay is an electrically operated switch. Many relays use an electromagnet to mechanically operate a switch. The advantage of relays is that it takes a relatively small amount of power to operate the relay coil, but the relay itself can be used to control motors, heaters, lamps or AC circuits which themselves can draw a lot more electrical power

6. **Solenoid**:



A solenoid is simply a specially designed electromagnet. Solenoids are inexpensive, and their use is primarily limited to on-off applications such as latching, locking, and triggering. They are frequently used in home appliances (e.g. washing machine valves), office equipment (e.g. copy machines), automobiles (e.g. door latches and the starter solenoid), pinball machines (e.g., plungers and bumpers), and factory automation